For GNSS setup:

```
File  Edit  Tabs  Help
pi@raspberrypi:~ $ source gnss_env/bin/activate
(gnss_env) pi@raspberrypi:~ $ sudo apt update
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
Hit:4 http://archive.raspberrypi.com/debian trixie InRelease
218 packages can be upgraded. Run 'apt list --upgradable' to see them.
(gnss_env) pi@raspberrypi:~ $ sudo apt install usb-modeswitch usb-modeswitch-data
usb-modeswitch is already the newest version (2.6.1-4+b2).
usb-modeswitch-data is already the newest version (20191128-7).
usb-modeswitch-data set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 218
(gnss_env) pi@raspberrypi:~ $ sudo modprobe option
(gnss_env) pi@raspberrypi:~ $ sudo modprobe usbserial
(gnss_env) pi@raspberrypi:~ $ echo "2c7c 0904" | sudo tee /sys/bus/usb-serial/drivers/option1/new_id
2c7c 0904
(gnss_env) pi@raspberrypi:~ $ dmesg | tail -20
[  993.663068] usb 1-1.3: config 1 has an invalid interface number: 8 but max is 7
[  993.663090] usb 1-1.3: config 1 has no interface number 6
[  993.663108] usb 1-1.3: config 1 interface 0 altsetting 0 endpoint 0x81 has an invalid bInterval 32, changing to 9
[  993.663502] usb 1-1.3: New USB device found, idVendor=2c7c, idProduct=0904, bcdDevice= 3.18
[  993.663522] usb 1-1.3: New USB device strings: Mfr=1, Product=2, SerialNumber=3
[  993.663535] usb 1-1.3: Product: Android
[  993.663546] usb 1-1.3: Manufacturer: Android
[  993.675039] cdc_ether 1-1.3:1.0 usb0: register 'cdc_ether' at usb-0000:01:00.0-1.3, CDC Ethernet Device, 02:4b:b3:b9:eb:e5
[  993.677528] option 1-1.3:1.2: GSM modem (1-port) converter detected
[  993.678388] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB0
[  993.679795] option 1-1.3:1.3: GSM modem (1-port) converter detected
[  993.680808] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB1
[  993.682221] option 1-1.3:1.4: GSM modem (1-port) converter detected
[  993.682624] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB2
[  993.683635] option 1-1.3:1.5: GSM modem (1-port) converter detected
[  993.684128] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB3
[  993.684916] option 1-1.3:1.7: GSM modem (1-port) converter detected
[  993.685310] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB4
[  993.686149] option 1-1.3:1.8: GSM modem (1-port) converter detected
[  993.686511] usb 1-1.3: GSM modem (1-port) converter now attached to ttyUSB5
(gnss_env) pi@raspberrypi:~ $ ls /dev/ttyUSB*
/dev/ttyUSB0  /dev/ttyUSB2  /dev/ttyUSB4
/dev/ttyUSB1  /dev/ttyUSB3  /dev/ttyUSB5
(gnss_env) pi@raspberrypi:~ $
```

AT commands:

```
Welcome to minicom 2.10

OPTIONS: I18n
Port /dev/ttyUSB0, 17:35:23 [U]

Press CTRL-A Z for help on special keys


OK

OK

+QGPSGNMEA: $GPGSV,3,1,11,41,75,167,44,15,69,088,45,29,68,353,41,45,67,135,43,1*6E
+QGPSGNMEA: $GPGSV,3,2,11,40,57,245,34,20
OK

+QGPSLOC: 120846.000,13.01070,80.23495,1.2,6.0,3,000.00,0.0,0.0,170326,15
```

AT commands:

AT

Turn ON GNSS: AT+QGPS=1

Check satellites: AT+QGPSGNMEA="GSV"

Get location: AT+QGPSLOC=2

GNSS Program:

```
import serial
import requests
import time

# -------------------------------
# Convert NMEA → Decimal Degrees
# -------------------------------
def convert_to_decimal(coord, direction):
    if not coord:
        return None

    try:
        if direction in ['N', 'S']:
            degrees = int(coord[:2])
            minutes = float(coord[2:])
        else:
            degrees = int(coord[:3])
            minutes = float(coord[3:])

        decimal = degrees + (minutes / 60)

        if direction in ['S', 'W']:
            decimal = -decimal

        return decimal
    except:
        return None


# -------------------------------
# Reverse Geocoding
# -------------------------------
def get_location_full(lat, lon):
    url = f"https://nominatim.openstreetmap.org/reverse?lat={lat}&lon={lon}&format=json"

    headers = {
        "User-Agent": "RaspberryPi-GNSS-Project"
    }

    try:
```

```
        response = requests.get(url, headers=headers, timeout=5)
        data = response.json()

        return data.get("display_name", "")

    except:
        return "Location fetch error"


# ------------------------------
# Extract Important Place
# ------------------------------
def extract_place(display_name):
    if not display_name:
        return "Unknown location"

    parts = display_name.split(',')

    # Look for important keywords
    keywords = ["University", "College", "School", "Hospital", "Temple", "Mall"]

    for part in parts:
        for key in keywords:
            if key.lower() in part.lower():
                return part.strip()

    # fallback → first part
    return parts[0].strip()


# ------------------------------
# MAIN PROGRAM
# ------------------------------
ser = serial.Serial('/dev/ttyUSB1', 115200, timeout=1)

print("Reading GNSS data...\n")

last_location_full = "Fetching..."
last_api_time = 0

while True:
    try:
        line = ser.readline().decode('utf-8', errors='ignore')

        if "$GNRMC" in line:
            parts = line.split(',')

            if parts[2] == 'A':  # Valid fix
                lat = parts[3]
                lat_dir = parts[4]
                lon = parts[5]
```

```
        lon_dir = parts[6]

        lat_dec = convert_to_decimal(lat, lat_dir)
        lon_dec = convert_to_decimal(lon, lon_dir)

        if lat_dec and lon_dec:

            # Update location every 5 seconds
            if time.time() - last_api_time > 5:
                last_location_full = get_location_full(lat_dec, lon_dec)
                last_api_time = time.time()

            # Extract clean place name
            location_clean = extract_place(last_location_full)

            # Print output
            print(f"Latitude : {lat_dec:.6f}")
            print(f"Longitude: {lon_dec:.6f}")
            print("Place    :", location_clean)
            print("Full Loc :", last_location_full)
            print("-" * 50)

except KeyboardInterrupt:
    print("\nStopped by user")
    ser.close()
    break

except Exception as e:
    print("Error:", e)
```

For LTE Setup:

```
pi@raspberrypi:~ $ sudo systemctl enable ModemManager
pi@raspberrypi:~ $ sudo systemctl start ModemManager
pi@raspberrypi:~ $ sudo systemctl enable NetworkManager
pi@raspberrypi:~ $ sudo systemctl start NetworkManager
pi@raspberrypi:~ $ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 2109:3431 VIA Labs, Inc. Hub
Bus 001 Device 010: ID 2c7c:0904 Quectel Wireless Solutions Co., Ltd. Android
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
pi@raspberrypi:~ $ sudo modprobe option
pi@raspberrypi:~ $ sudo modprobe usb_wwan
pi@raspberrypi:~ $ echo "2c7c 0904" | sudo tee /sys/bus/usb-serial/drivers/option1/new-id
tee: /sys/bus/usb-serial/drivers/option1/new-id: Permission denied
2c7c 0904
pi@raspberrypi:~ $ ls /dev/ttyUSB*
/dev/ttyUSB0  /dev/ttyUSB2  /dev/ttyUSB4
/dev/ttyUSB1  /dev/ttyUSB3  /dev/ttyUSB5
pi@raspberrypi:~ $ sudo mmcli -L
    /org/freedesktop/ModemManager1/Modem/0 [Quectel] EC200G
pi@raspberrypi:~ $ nmcli con show
NAME              UUID                                  TYPE      DEVICE
netplan-wlan0-pi  acfb25b1-cbcd-3c83-8ca6-15f2af0ab563  wifi      wlan0
lte               2dd7f195-c037-4274-93a1-20695d17d8eb  gsm       ttyUSB5
lo                e48d9338-1464-4748-a114-8e4782c1d0e5  loopback  lo
netplan-eth0      75a1216a-9d1a-30cd-8aca-ace5526ec021  ethernet  --
pi@raspberrypi:~ $ sudo nmcli con up lte
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/7)
pi@raspberrypi:~ $ ping -c 4 google.com
PING google.com (2404:6800:4007:804::200e) 56 data bytes
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=1 ttl=116 time=27.8 ms
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=2 ttl=116 time=25.3 ms
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=3 ttl=116 time=57.1 ms
64 bytes from pnmaaa-ap-in-x0e.1e100.net (2404:6800:4007:804::200e): icmp_seq=4 ttl=116 time=54.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 25.321/41.234/57.061/14.703 ms
pi@raspberrypi:~ $
```

AT Commands:

```
+CGREG: 2,1,"B049","A014D34",7

OK

+CME ERROR: 58

+CSQ: 26,99

+QNWINFO: "TDD LTE","40440","LTE BAND 40",39150

OK

OK

OK

OK

OK

OK

+QIACT: 1,1,1,"100.93.183.33"
```

AT commands:

Check SIM status: AT+CPIN?

Check network registration: AT+CREG? , AT+CGREG?

Check signal strength: AT+CSQ

Check network type: AT+QNWINFO

Set APN: AT+CGDCONT=1,"IP","airtelgprs.com"

Activate data: AT+CGATT=1 , AT+QIACT=1

Check IP address: AT+QIACT?